

Tutorial 2, Friday, 09/10/2026, by vrm

Today: *LTL*, *CTL*, and *CTL****A. Automata and Expressions for Infinite Words****Büchi Automata.** A *nondeterministic automaton* is a tuple

$$\mathcal{A} = (Q, \Sigma, \delta, q_0, \mathcal{F}),$$

where Q is a finite set of states, Σ is a finite alphabet, $\delta \subseteq Q \times \Sigma \times Q$ is the transition relation, $q_0 \in Q$ is the initial state, and $\mathcal{F}$ is some sort of *acceptance condition*. For a run

$$\rho : q_0 \xrightarrow{a_0} q_1 \xrightarrow{a_1} q_2 \xrightarrow{a_2} \dots$$

we shall write

$$\text{Inf}(\rho) = \{q \in Q : q = q_i \text{ for infinitely many } i\}.$$

In a *Büchi automaton* (NBA), the acceptance condition is generated by a set $F \subseteq Q$ of states, and a run ρ is *accepting* if it visits F infinitely often: $\text{Inf}(\rho) \cap F \neq \emptyset$.

Other automata. A *Muller automaton* is equipped with an acceptance family $\mathcal{F} \subseteq 2^Q$. A run ρ is accepting if $\text{Inf}(\rho) \in \mathcal{F}$.

Similarly, a *Rabin automaton* is an automaton equipped with an acceptance condition

$$\mathcal{R} = \{(E_1, F_1), \dots, (E_k, F_k)\}, \quad E_i, F_i \subseteq Q.$$

A run ρ is accepting if there exists $i \in \{1, \dots, k\}$ such that

$$\text{Inf}(\rho) \cap E_i = \emptyset \quad \text{and} \quad \text{Inf}(\rho) \cap F_i \neq \emptyset,$$

i.e. if it visits an “allowed” set E_i infinitely often, and avoids a “forbidden” set co-finitely often (co-finite: always, except a finite number of times).

Finally, a *parity automaton* is an automaton equipped with a priority function

$$\Omega : Q \longrightarrow \{0, \dots, d\},$$

and a run ρ is accepting if

$$\min\{\Omega(q) : q \in \text{Inf}(\rho)\}$$

is even.

ω -Regular Languages An ω -regular expression over an alphabet Σ is defined inductively by the grammar

$$s ::= r^\omega \mid rs \mid s + s,$$

where r ranges over *regular expressions* over Σ . Its language $\mathcal{L}(s) \subseteq \Sigma^\omega$ is defined inductively by

$$\mathcal{L}(r^\omega) = \mathcal{L}(r)^\omega,$$

$$\mathcal{L}(rs) = \mathcal{L}(r)\mathcal{L}(s),$$

$$\mathcal{L}(s_1 + s_2) = \mathcal{L}(s_1) \cup \mathcal{L}(s_2).$$

A language $L \subseteq \Sigma^\omega$ is called ω -regular if $L = \mathcal{L}(s)$ for some ω -regular expression s .

B. Exercises

Exercise 1. [*Constructing Büchi automata*] Let $\Sigma = \{a, b\}$. Construct Büchi automata recognizing the following languages.

1. Words containing infinitely many occurrences of a .
2. Words containing only finitely many occurrences of a .
3. Words containing infinitely many occurrences of both a and b .
4. Words in which every occurrence of a is eventually followed by an occurrence of b .
5. Words containing infinitely many occurrences of the factor ab .
6. Words that are ultimately periodic with period ab :

$$\{u(ab)^\omega : u \in \Sigma^*\}.$$

7. Words containing infinitely many a 's, but no two consecutive occurrences of a .
8. Words having some suffix belonging to $(a^*b)^\omega$.

Which of these languages can be recognized by deterministic Büchi automata?

Exercise 2. [*Pumping lemma for Büchi automata*] Let $L \subseteq \Sigma^\omega$ be recognized by a Büchi automaton with n states. For every $w \in L$, there exist $x, y \in \Sigma^*$ and $z \in \Sigma^\omega$ such that

$$w = xyz, \quad |xy| \leq n, \quad |y| > 0,$$

and

$$xy^kz \in L \quad \text{for every } k \geq 0.$$

Proof. Let $w = a_0a_1a_2 \cdots \in L$, and consider an accepting run

$$q_0 \xrightarrow{a_0} q_1 \xrightarrow{a_1} q_2 \xrightarrow{a_2} \cdots.$$

Among the first $n + 1$ states $q_0, \dots, q_n$, two must coincide. Thus there exist $0 \leq i < j \leq n$ such that $q_i = q_j$.

Write

$$x = a_0 \cdots a_{i-1}, \quad y = a_i \cdots a_{j-1}, \quad z = a_j a_{j+1} \cdots.$$

Since $q_i = q_j$, the portion of the run reading y is a cycle. Consequently, we may traverse this cycle any finite number of times, obtaining a run on xy^kz for every $k \geq 0$.

The resulting run is accepting, since it differs from the original accepting run only by a finite prefix. Hence $xy^kz \in L$ for every $k \geq 0$. $\square$

Exercise 3. [*Non- ω -regular languages*] Use the pumping lemma to prove that the following languages are not ω -regular.

- 1.

$$L_1 = \{a^n b^n c^\omega : n \geq 1\}.$$

- 2.

$$L_2 = \{a^n b^{2n} (ab)^\omega : n \geq 1\}.$$

3.

$$L_3 = \{a^n b^m c^\omega : n, m \geq 1, n < m < 2n\}.$$

4.

$$L_4 = \{a^{n_1} b a^{n_2} b a^{n_3} b \dots : n_i = n_1 \text{ for every } i \geq 1\}.$$

Exercise 4. Every nonempty language recognized by a Büchi automaton contains an ultimately periodic word, i.e., a word of the form

$$uv^\omega, \quad u, v \in \Sigma^*, \quad v \neq \varepsilon.$$

Proof. Consider an accepting run. Some accepting state occurs infinitely often. Choose two occurrences of that state and repeat the finite segment of the run between them forever. $\square$

Exercise 5. [*Languages without ultimately periodic words*] Prove that the following languages are not ω -regular.

1. The language of infinite words over $\{a, b\}$ containing infinitely many b 's and arbitrarily long consecutive blocks of a 's.

2.

$$L = \{a^{n_1} b a^{n_2} b a^{n_3} b \dots : n_1 < n_2 < n_3 < \dots\}.$$

1. Other Acceptance Conditions

Exercise 6. [*Acceptance conditions*] Let $Q = \{p, q, r\}$. Describe which runs are accepted by each of the following conditions.

1. The Muller condition

$$\mathcal{F} = \{\{p\}, \{p, q\}, \{q, r\}\}.$$

2. The Rabin condition

$$\mathcal{R} = \{(\{p\}, \{q\}), (\{q\}, \{r\})\}.$$

3. The parity condition

$$\Omega(p) = 0, \quad \Omega(q) = 1, \quad \Omega(r) = 2.$$

4. Give a Muller condition equivalent to the Rabin condition in (b).
5. Give a Rabin condition equivalent to the parity condition in (c).

2. Equivalence of Acceptance Conditions

Exercise 7. [*Büchi acceptance as other acceptance conditions*] Let $\mathcal{A}$ be a Büchi automaton with accepting set F . Show that its acceptance condition can be expressed as follows.

1. As a Muller condition:

$$\mathcal{F} = \{S \subseteq Q : S \cap F \neq \emptyset\}.$$

2. As a Rabin condition consisting of the single pair

$$(\emptyset, F).$$

3. As a parity condition:

$$\Omega(q) = \begin{cases} 0, & q \in F, \\ 1, & q \notin F. \end{cases}$$

Exercise 8. [*Parity to Rabin*] Let $\mathcal{A}$ be a parity automaton with priority function

$$\Omega : Q \rightarrow \{0, \dots, d\}.$$

For every even priority $2i \leq d$, define

$$E_i = \{q : \Omega(q) < 2i\}, \quad F_i = \{q : \Omega(q) = 2i\}.$$

Prove that the Rabin condition consisting of the pairs (E_i, F_i) is equivalent to the original parity condition.

Exercise 9. [*Rabin to Muller*] Let

$$\mathcal{R} = \{(E_i, F_i) : 1 \leq i \leq k\}$$

be a Rabin acceptance condition. Define

$$\mathcal{F} = \{S \subseteq Q : \exists i (S \cap E_i = \emptyset \wedge S \cap F_i \neq \emptyset)\}.$$

Prove that the Muller condition $\mathcal{F}$ is equivalent to $\mathcal{R}$.

Exercise 10. [*Rabin to Büchi*] Show that every language recognized by a nondeterministic Rabin automaton is recognized by a nondeterministic Büchi automaton.

Hint. Nondeterministically choose a Rabin pair (E_i, F_i) and guess a position after which no state from E_i is visited. From that position onward, require infinitely many visits to F_i .

Exercise 11. [*Muller to Büchi*] Show that every language recognized by a nondeterministic Muller automaton is recognized by a nondeterministic Büchi automaton.

Hint. Nondeterministically select $S \in \mathcal{F}$, guess a position after which the run visits only states of S , and ensure that every state of S occurs infinitely often.

Exercise 12. The following classes of languages coincide:

$$\text{NBA} = \text{NPA} = \text{NRA} = \text{NMA}.$$

Proof. The preceding exercises establish

$$\text{NBA} \subseteq \text{NPA} \subseteq \text{NRA} \subseteq \text{NMA} \subseteq \text{NBA}.$$

□

Remark! The analogous equivalence does not hold for deterministic Büchi automata. For example,

$$L = \{w \in \{a, b\}^\omega : a \text{ occurs only finitely often}\}$$

is recognized by a deterministic parity automaton, but not by any deterministic Büchi automaton.

Büchi's characterization For every $L \subseteq \Sigma^\omega$, the following are equivalent:

1. L is ω -regular.
2. L is recognized by a nondeterministic Büchi automaton.

Exercise 13. [*Büchi's characterization*] Prove the preceding theorem.

1. Let $U, V \subseteq \Sigma^*$ be regular languages, with $\varepsilon \notin V$. Construct a Büchi automaton recognizing

$$UV^\omega.$$

Hint. Combine finite automata for U and V . The Büchi automaton should read a prefix in U , then repeatedly read nonempty words in V , visiting an accepting state after each repetition.

Explain how to extend the construction to finite unions.

2. Let

$$\mathcal{A} = (Q, \Sigma, \delta, q_0, F)$$

be a Büchi automaton. For $p, q \in Q$, define

$$L_{p,q} = \{u \in \Sigma^* : \text{there exists a path from } p \text{ to } q \text{ labeled } u\}.$$

Observe that each $L_{p,q}$ is regular.

Prove that

$$\mathcal{L}_\omega(\mathcal{A}) = \bigcup_{q \in F} L_{q_0,q} (L_{q,q} \setminus \{\varepsilon\})^\omega.$$

Hint. In an accepting run, some accepting state occurs infinitely often. Split the run at successive occurrences of that state.

3. Decision Problems

The fundamental decision problems for Büchi automata are:

1. *Emptiness*: given $\mathcal{A}$, decide whether

$$\mathcal{L}_\omega(\mathcal{A}) = \emptyset.$$

2. *Intersection emptiness*: given $\mathcal{A}, \mathcal{B}$, decide whether

$$\mathcal{L}_\omega(\mathcal{A}) \cap \mathcal{L}_\omega(\mathcal{B}) = \emptyset.$$

3. *Universality*: given $\mathcal{A}$, decide whether

$$\mathcal{L}_\omega(\mathcal{A}) = \Sigma^\omega.$$

Exercise 14. [*Emptiness*] The emptiness problem for Büchi automata is decidable.

More precisely, a Büchi automaton accepts some word if and only if its transition graph contains a reachable directed cycle passing through an accepting state.

What is the complexity of deciding Emptiness of Büchi automata? [Hint: linear time in the size of the transition graph.]

Proof. Suppose $\mathcal{A}$ accepts some word, and let ρ be an accepting run. Some accepting state $q \in F$ appears infinitely often in ρ . Hence q is reachable from the initial state, and there exists a nonempty path from q back to q .

Conversely, suppose there is a path from q_0 to an accepting state q , together with a nonempty path from q back to q . Let u and v label these paths, respectively. Then

$$uv^\omega$$

is accepted by $\mathcal{A}$. □

Exercise 15. [*Emptiness algorithm*] Design an algorithm deciding whether a given Büchi automaton has a nonempty language.

1. Describe an algorithm based on strongly connected components.
2. Prove its correctness.
3. Analyze its time complexity.
4. If the language is nonempty, show how to construct an accepted word of the form uv^ω .

Exercise 16. [*Intersection*] Let $\mathcal{A}_1, \mathcal{A}_2$ be Büchi automata.

Prove that there exists a Büchi automaton $\mathcal{B}$ such that

$$\mathcal{L}_\omega(\mathcal{B}) = \mathcal{L}_\omega(\mathcal{A}_1) \cap \mathcal{L}_\omega(\mathcal{A}_2).$$

Hint. The naive synchronous product with accepting set $F_1 \times F_2$ does not work. Add a component that alternates between waiting for a visit to F_1 and waiting for a visit to F_2 .

1. Generalize the construction to the intersection of k Büchi languages.
2. Deduce that intersection emptiness is decidable.
3. Determine the size of your construction.

Exercise 17. [*Universality*] Consider the universality problem for nondeterministic Büchi automata:

$$\mathcal{L}_\omega(\mathcal{A}) = \Sigma^\omega ?$$

1. Explain why the usual subset construction for finite automata does not immediately solve the problem.
2. Assuming that Büchi automata are effectively closed under complementation, prove that universality is decidable.
3. Show that universality is decidable for complete deterministic Büchi automata without constructing a Büchi automaton for the complement.

Hint. A deterministic Büchi automaton rejects a word if its unique run visits the accepting states only finitely often.

4. Closure Under Complementation

Exercise 18. [*Complementing Büchi automata*] For finite words, a complete deterministic finite automaton can be complemented by replacing its accepting set F with $Q \setminus F$.

Show that this construction does not generally work for deterministic Büchi automata.

Hint. Consider a run visiting both F and $Q \setminus F$ infinitely often.

Exercise 19. [*Complementing deterministic Muller automata*] Let

$$\mathcal{A} = (Q, \Sigma, \delta, q_0, \mathcal{F})$$

be a complete deterministic Muller automaton.

Define

$$\bar{\mathcal{F}} = 2^Q \setminus \mathcal{F}.$$

Prove that the automaton

$$\bar{\mathcal{A}} = (Q, \Sigma, \delta, q_0, \bar{\mathcal{F}})$$

recognizes the complement of $\mathcal{L}_\omega(\mathcal{A})$.

Exercise 20. [*Complementing deterministic parity automata*] Let $\mathcal{A}$ be a complete deterministic parity automaton with priority function

$$\Omega : Q \rightarrow \mathbb{N}.$$

Define

$$\Omega'(q) = \Omega(q) + 1.$$

Prove that replacing Ω with Ω' gives an automaton recognizing

$$\Sigma^\omega \setminus \mathcal{L}_\omega(\mathcal{A}).$$

McNaughton's determinization theorem Every nondeterministic Büchi automaton has an equivalent deterministic Muller automaton.

Closure under complementation If $L \subseteq \Sigma^\omega$ is ω -regular, then

$$\Sigma^\omega \setminus L$$

is ω -regular.

Proof. By Büchi's characterization, there exists a nondeterministic Büchi automaton $\mathcal{A}$ recognizing L .

By McNaughton's theorem, there exists an equivalent deterministic Muller automaton $\mathcal{B}$.

Complementing the Muller acceptance family of $\mathcal{B}$ produces a deterministic Muller automaton recognizing $\Sigma^\omega \setminus L$.

Finally, every language recognized by a Muller automaton is recognized by a nondeterministic Büchi automaton. Hence $\Sigma^\omega \setminus L$ is ω -regular. $\square$

5. Further Exercises

Exercise 21. [*Boolean closure*] Prove that the class of ω -regular languages is closed under

$$\cup, \quad \cap, \quad \setminus,$$

and symmetric difference.

Exercise 22. [*Finite prefixes*] Let $L \subseteq \Sigma^\omega$ be ω -regular and $K \subseteq \Sigma^*$ be regular. Prove that

$$KL = \{uv : u \in K, v \in L\}$$

is ω -regular.

Exercise 23. [*Prefix independence*] Call an ω -language L *prefix-independent* if

$$uw \in L \iff w \in L$$

for every $u \in \Sigma^*$ and $w \in \Sigma^\omega$.

1. Prove that the following languages are prefix-independent:

- (a) words containing infinitely many a 's;
- (b) words containing finitely many a 's;
- (c) words containing infinitely many occurrences of ab .

2. Give an example of an ω -regular language that is not prefix-independent.

3. Show that a language recognized by a complete deterministic Muller, Rabin, or parity automaton need not be prefix-independent.

Exercise 24. [*Ultimately periodic words*] Let

$$\text{UP} = \{uv^\omega : u, v \in \Sigma^*, v \neq \varepsilon\}.$$

Prove that two ω -regular languages $L_1, L_2 \subseteq \Sigma^\omega$ are equal if and only if

$$L_1 \cap \text{UP} = L_2 \cap \text{UP}.$$

Hint. If $L_1 \neq L_2$, their symmetric difference is a nonempty ω -regular language. Apply the proposition asserting the existence of ultimately periodic words.